A twenty-seven-variable candidate equation

Positive integer unknowns; exact degree 8; a 129-operation evaluator.

The only unknowns are the twenty-seven separately named positive-integer variables

$$J, F, \alpha, f, g, T, X, Y, \xi, \eta, \zeta, \upsilon, \gamma, \psi, \varphi, \epsilon,$$
$$c, \ell, n, Z, D, V, \delta,$$
$$\kappa, \lambda, \theta, Q \in \mathbb{Z}_{>0}.$$

All other symbols below are polynomial abbreviations:

$$b = \beta J, \quad q = 1 + b, \quad C = q - F - \alpha, \quad U = Y\kappa,$$
$$a = \lambda + Y, \quad s = q\kappa, \quad x = \lambda q,$$
$$W = C(X + Kq) - Fq, \quad m = f^2,$$
$$\sigma = \psi^2, \quad \nu = \upsilon^2,$$
$$E = \lambda, \quad k = \eta + \zeta,$$
$$\tau = \ell(\ell + k), \quad \chi = \varphi^2,$$
$$H = 3 + 4a, \quad \Delta = \delta, \quad L = X + ac,$$
$$M = sb,$$
$$R = [\alpha + (q - F)b](\kappa - 1) + J(\mu + \omega q),$$
$$\vartheta = V^2,$$
$$I = \theta,$$
$$\mathcal{P} = Z^2 - \sigma(1 + \tau),$$
$$\mathcal{D} = D^2 - \chi(1 + \Delta n),$$
$$\mathcal{Q} = \nu - Q[\nu - \vartheta] - 1,$$
$$\mathcal{R} = I^2 - Q,$$
$$\mathcal{B} = \psi - M, \quad \mathcal{C} = \varphi - U, \quad \mathcal{G} = \epsilon - R,$$
$$\mathcal{H} = c - \eta - Yk,$$
$$\mathcal{L} = \ell - EYk,$$
$$\mathcal{N} = n - c^2,$$
$$\mathcal{I} = Z - [\psi(\ell + k + \xi - kW) + Yb - Ws],$$
$$\mathcal{J} = D - [\varphi L + H(\varphi\gamma + x + \epsilon + 1 - k)],$$
$$\mathcal{K} = V - [m(Tc - \epsilon) - c], \quad \mathcal{O} = \delta - H - a^2,$$
$$\mathfrak{J} = \kappa - q^2, \quad \mathcal{F} = \lambda - XY,$$
$$\mathcal{V} = \theta - ng,$$
$$\mathfrak{B} = Q - \Delta(m - 1).$$

The single Diophantine equation is

$$\mathcal{P}^2 + \mathcal{D}^2 + \mathcal{Q}^2 + \mathcal{R}^2 + \mathcal{B}^2 + \mathcal{C}^2 + \mathcal{G}^2 + \mathcal{H}^2 + \mathcal{L}^2 + \mathcal{N}^2 + \mathcal{I}^2 + \mathcal{J}^2 + \mathcal{K}^2 + \mathcal{O}^2 + \mathfrak{J}^2 + \mathcal{F}^2 + \mathcal{V}^2 + \mathfrak{B}^2 = 0.$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{2011}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1005}.$$

$$\mu = \beta - G_{951}(\rho) - \rho^{960}G_{26}(\rho) - \rho^{1004},$$
$$\omega = \beta + 8G_{917}(\rho) - 4 + (2^{45} - 4)\Lambda + \rho^{960}G_9(\rho) + \rho^{987}G_8(\rho) + \rho^{1004},$$
$$K = \rho^{18} + \rho^{1004} + \rho^{2020} + 2\Lambda\Phi(1/\rho),$$

where

$$\chi_i = 1 + 2^{10}(1 + w_i + 2^8 n_i + 2^{17}p_i + 2^{25}s_i) \quad (0 \le i \le 941),$$
$$\pi(0) = 0, \quad \pi(i) = 1 + \sum_{j=1}^{941} \left\lfloor \frac{2^{44} - 1 + \chi_j - \chi_i}{2^{44}} \right\rfloor \quad (1 \le i \le 941).$$

$$\Phi(x) = \sum_{i=0}^{941} \chi_i x^{\pi(i)} + 2^{10}x^{969}(x^9 + 2^{17})G_8(2x) + 2^{18}x^{942}(x^{18} + 2^{17})G_9(2x).$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q - 1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j + 1$ $(j = 2q + b < 430)$	$((1 - d_j)r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{423+2r_{d,q}+b \\ q\in\mathcal{S}_d,\ d,b=0,1}$	$(dr_{d,q}, b, (1 - d)r_{d,q}, h(q, b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\mathcal{N} = \mathcal{A} \bmod 2^{636}, \quad \mathcal{T} = \lfloor \mathcal{A}/2^{636} \rfloor,$$
$$\nu_j = D_2(\mathcal{N}, j), \quad \kappa_j = (\sigma_j + 2)^{1-\nu_j} \quad (0 \le j < 430),$$
$$\sigma_j = \sum_{k=0}^{j-1} \nu_k, \quad \Pi_j = \prod_{k=0}^{j-1} \kappa_k \quad (0 \le j \le 430),$$
$$m_q = D_2(\mathcal{N}, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(\lfloor \mathcal{T}/\Pi_{430} \rfloor, \tau_q) - 14),$$
$$u_j = (\sigma_{j+1} - (\lfloor \mathcal{T}/\Pi_j \rfloor \bmod \kappa_j)) \bmod 216,$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums/products are 0/1. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$ in base 10. Its complete decimal expansion is the concatenation of these lines:

```
12889260410359561746736540743515845134244260640012702033212621
95398074180784058571518488282119861023093355389406939414042892
97455773992193784795705001618777025966369137954859764532832722
75411843410809038713502633204617333246733616940964578272360901
56186216508504953649192752154519268024509952082950238865468307
22999632482769253980274091217460707829023970817244790340704449
09085581271192071656170732004937337082171141082486384269072884
72778087419663417942892520175630649792329265834490253035577533
96449689808375788457488971017043689875433162972390177936123032
89025657757489526342497438772045714099661532481123050235898973
27443609903800541199351405985754278307227777584454508422717998
33411356440933534607489268032843 7
```

The fixed integer $\mathcal{A}$ above is written in base 10. This equation is a candidate: its model witness transport and native intuitionistic PRA equivalence have not yet been proved in Lean. Its exact degree is certified by integer-polynomial calculations; the 129-operation count is for the unshifted circuit. Neither count is a minimality claim.

Exact 129-operation circuit

Each numbered line defines one gate over the integers. The inputs are the twenty-seven unknowns and the four fixed integers above. The ASCII names `alpha, xi, eta, zeta, upsilon, gamma, vartheta, psi, varphi, epsilon, beta, mu, omega, ell, Delta, delta, kappa, lambda, sigma, tau, chi, theta, nu` refer to their displayed counterparts; `g` is the unknown g. The last gate is zero exactly when the displayed equation holds. The circuit inputs `cNamed, ellNamed` are c, ℓ; gate `a` is the displayed abbreviation; gate `WComputed` is the displayed abbreviation $W = C(X + Kq) - Fq$, and gate `m` is $m = f^2$. The inputs `n, Z, D, V, delta` are the displayed n, Z, D, V, δ; `kappa, lambda, theta, Q` are $\kappa, \lambda, \theta, Q$. Gates `q3, g55, g39, y2, g80` compute $s, x, \sigma, \nu, \vartheta$; gates `g40, g62` compute τ, χ. Gates `E, I` compute XY, ng; the display uses $E = \lambda$, $I = \theta$. Intermediate gate names below are local abbreviations, not additional unknowns.

```
1.  b = J * beta
2.  q = 1 + b
3.  q2 = q * q
4.  q3 = q * kappa
5.  U = Y * kappa
6.  qF = q - F
7.  C = qF - alpha
8.  E = X * Y
9.  k = eta + zeta
10.  kY = Y * k
11.  c = eta + kY
12.  c2 = cNamed * cNamed
13.  a = lambda + Y
14.  g15 = 4 * a
15.  H = 3 + g15
16.  g17 = a * a
17.  DeltaComputed = H + g17
18.  g19 = a * cNamed
19.  L = X + g19
20.  ell = lambda * kY
21.  M = q3 * b
22.  g25 = K * q
23.  g26 = X + g25
24.  g27 = C * g26
25.  g28 = F * q
26.  WComputed = g27 - g28
27.  g30 = ellNamed + k
28.  g31 = g30 + xi
29.  root_skew = k * WComputed
30.  root_inner = g31 - root_skew
31.  root_scaled = psi * root_inner
32.  g33 = Y * b
33.  root_btq3 = WComputed * q3
34.  root_sum = root_scaled + g33
35.  ZComputed = root_sum - root_btq3
36.  g38 = Z * Z
37.  g39 = psi * psi
38.  g40 = ellNamed * g30
39.  g41 = 1 + g40
40.  g42 = g39 * g41
41.  A0 = g38 - g42
42.  g44 = qF * b
43.  rankseed = alpha + g44
44.  g46 = kappa - 1
45.  g47 = rankseed * g46
46.  g48 = omega * q
47.  g49 = mu + g48
48.  g50 = J * g49
49.  R = g47 + g50
50.  g52 = k - epsilon
51.  r0 = g52 - 1
52.  g54 = varphi * gamma
53.  g55 = lambda * q
54.  g56 = g54 + g55
55.  shift = g56 - r0
56.  g58 = varphi * L
57.  g59 = H * shift
58.  DstarComputed = g58 + g59
59.  g61 = D * D
60.  g62 = varphi * varphi
61.  g63 = delta * n
62.  g64 = 1 + g63
63.  g65 = g62 * g64
64.  A1 = g61 - g65
65.  m = f * f
66.  rho = m - 1
67.  QComputed = delta * rho
68.  I = n * g
69.  g71 = theta * theta
70.  g72 = g71 - Q
71.  g74 = T * cNamed
72.  g75 = g74 - epsilon
73.  g76 = m * g75
74.  VComputed = g76 - cNamed
75.  y2 = upsilon * upsilon
76.  g80 = V * V
77.  g81 = y2 - g80
78.  g82 = Q * g81
79.  Fa = y2 - g82
80.  auxResidual = Fa - 1
81.  firstSquare = A0 * A0
82.  mainSquare = A1 * A1
83.  auxSquare = auxResidual * auxResidual
84.  strongSquare = g72 * g72
85.  firstMain = firstSquare + mainSquare
86.  firstMainAux = firstMain + auxSquare
87.  sumSquares = firstMainAux + strongSquare
88.  scaleDiff = psi - M
89.  scaleCheck = scaleDiff * scaleDiff
90.  final = sumSquares + scaleCheck
91.  mainDiff = varphi - U
92.  mainCheck = mainDiff * mainDiff
93.  result16 = final + mainCheck
94.  rankDiff = epsilon - R
95.  rankCheck = rankDiff * rankDiff
96.  result17 = result16 + rankCheck
97.  cDifference = cNamed - c
98.  cConstraint = cDifference * cDifference
99.  ellDifference = ellNamed - ell
100.  ellConstraint = ellDifference *
ellDifference
101.  addC = result17 + cConstraint
102.  addEll = addC + ellConstraint
103.  cSquareDifference = n - c2
104.  cSquareConstraint = cSquareDifference *
cSquareDifference
105.  addCSquare = addEll + cSquareConstraint
106.  zDifference = Z - ZComputed
107.  zConstraint = zDifference * zDifference
108.  dDifference = D - DstarComputed
109.  dConstraint = dDifference * dDifference
110.  vDifference = V - VComputed
111.  vConstraint = vDifference * vDifference
112.  deltaDifference = delta - DeltaComputed
113.  deltaConstraint = deltaDifference *
deltaDifference
114.  addZConstraint = addCSquare +
zConstraint
115.  addDConstraint = addZConstraint +
dConstraint
116.  addVConstraint = addDConstraint +
vConstraint
117.  addDELTAConstraint = addVConstraint +
deltaConstraint
118.  qSquareDifference = kappa - q2
119.  qSquareConstraint = qSquareDifference *
qSquareDifference
120.  product2Difference = lambda - E
121.  product2Constraint = product2Difference
* product2Difference
122.  scaledIndexDifference = theta - I
123.  scaledIndexConstraint =
scaledIndexDifference * scaledIndexDifference
124.  qSignedDifference = Q - QComputed
125.  qSignedConstraint = qSignedDifference *
qSignedDifference
126.  add_qSquare = addDELTAConstraint +
qSquareConstraint
127.  add_product2 = add_qSquare +
product2Constraint
128.  add_scaledIndex = add_product2 +
scaledIndexConstraint
129.  add_qSigned = add_scaledIndex +
qSignedConstraint
```